

Day 29. Anomaly detection

Anomaly detection

Anomaly detection example

Anomaly detection example

Aircraft engine features:

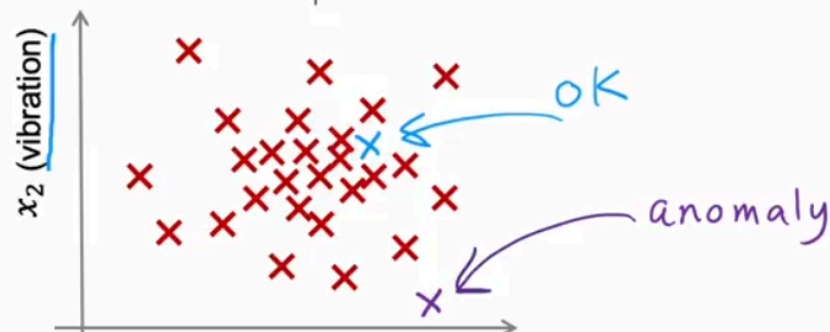
x_1 = heat generated

x_2 = vibration intensity

...

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

New engine: x_{test}



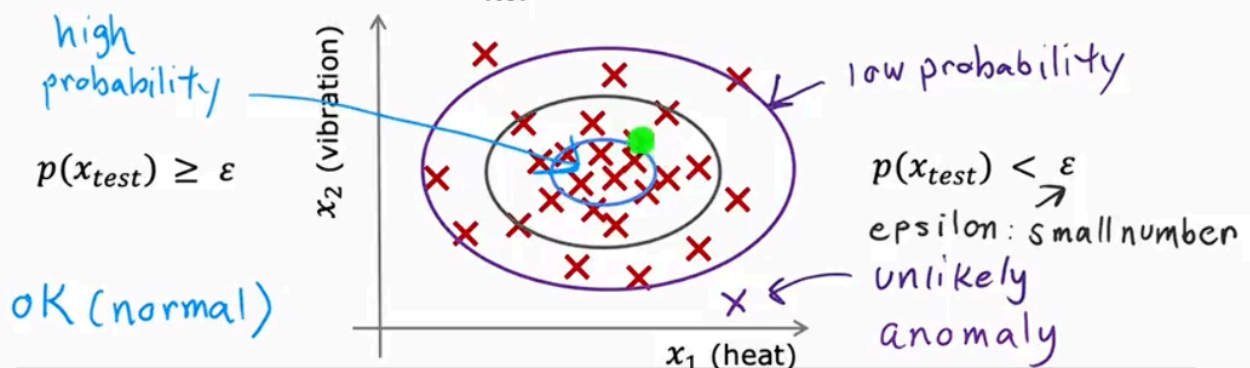
Density estimation

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

probability of x being seen in dataset

Model $p(x)$

Is x_{test} anomalous?



Anomaly detection example

Fraud detection:

- $x^{(i)}$ = features of user i 's activities
- Model $p(x)$ from data.
- Identify unusual users by checking which have $p(x) < \epsilon$

how often log in?

how many web pages visited?

transactions?

posts? typing speed?

perform additional checks to identify real fraud vs. false alarms

Manufacturing:

$x^{(i)}$ = features of product i

airplane engine

circuit board

smartphone

ratios

Monitoring computers in a data center:

$x^{(i)}$ = features of machine i

- x_1 = memory use,
- x_2 = number of disk accesses/sec,
- x_3 = CPU load,
- x_4 = CPU load/network traffic.

Gaussian (normal) distribution

Gaussian (Normal) distribution

σ standard deviation

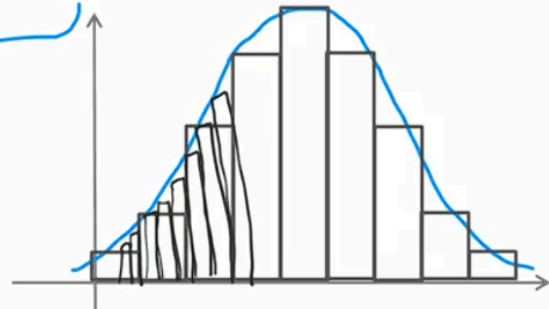
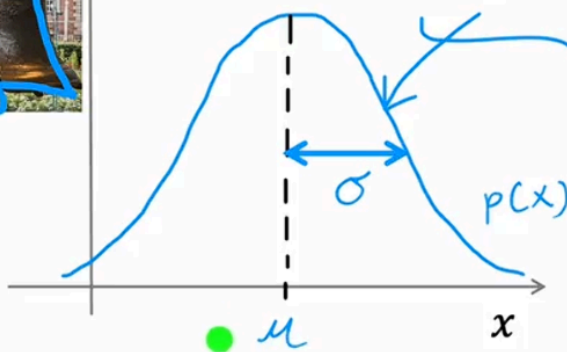
σ^2 variance

Say x is a number.

Probability of x is determined by a Gaussian with mean μ , variance σ^2 .

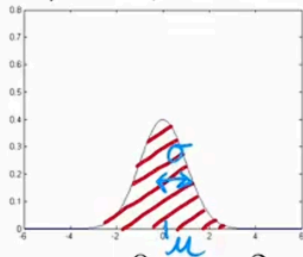
$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\pi = 3.14$

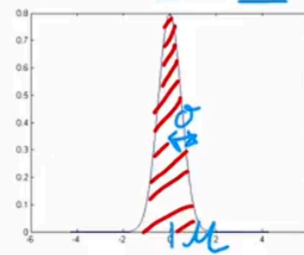


Gaussian distribution example

$\mu = 0, \sigma = 1$

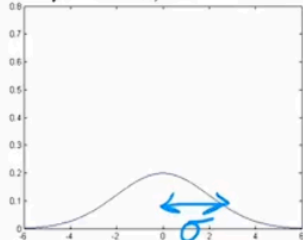


$\mu = 0, \sigma = 0.5$



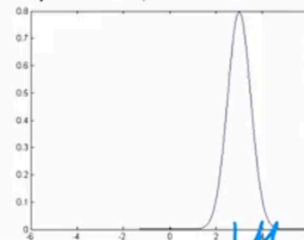
$\sigma^2 = 0.25$

$\mu = 0, \sigma = 2$



$\sigma^2 = 4$

$\mu = 3, \sigma = 0.5$



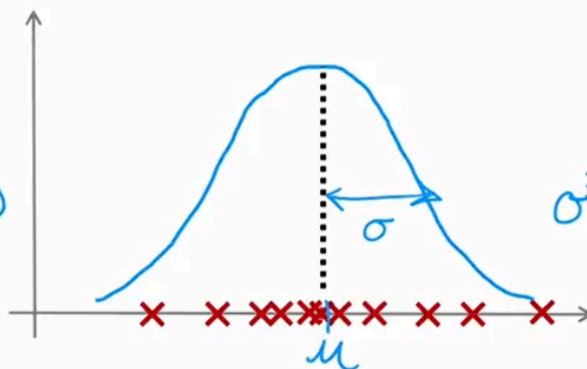
Parameter estimation

maximum likelihood

for μ, σ

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

$$\mu = \frac{1}{m} \sum_{i=1}^m x^{(i)}$$



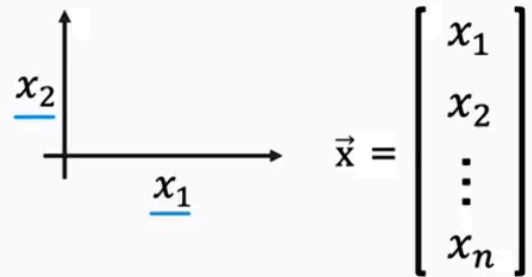
$\frac{1}{m-1}$

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x^{(i)} - \mu)^2$$

Anomaly detection algorithm

Density estimation

Training set: $\{\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(m)}\}$
 Each example $\vec{x}^{(i)}$ has n features



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$p(\vec{x}) = p(x_1; \mu_1, \sigma_1^2) * p(x_2; \mu_2, \sigma_2^2) * p(x_3; \mu_3, \sigma_3^2) * \dots * p(x_n; \mu_n, \sigma_n^2)$$

$$= \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2)$$

$\sum$ "add" $\prod$ "multiply"

$$p(x_1 = \text{high temp}) = 1/10$$

$$p(x_2 = \text{high vibra}) = 1/20$$

$$p(x_1, x_2) = p(x_1) * p(x_2)$$

$$= \frac{1}{10} * \frac{1}{20} = \frac{1}{200}$$

Anomaly detection algorithm

1. Choose n features x_i that you think might be indicative of anomalous examples.
2. Fit parameters $\mu_1, \dots, \mu_n, \sigma_1^2, \dots, \sigma_n^2$

$$\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)} \quad \sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (x_j^{(i)} - \mu_j)^2$$

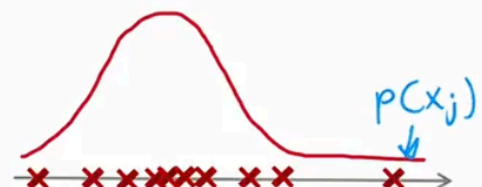
Vectorized formula

$$\vec{\mu} = \frac{1}{m} \sum_{i=1}^m \vec{x}^{(i)} \quad \vec{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

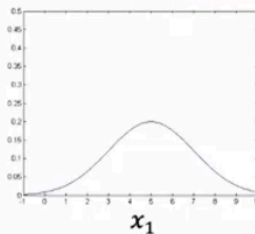
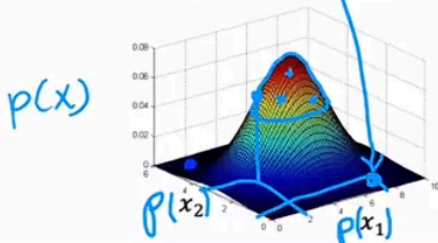
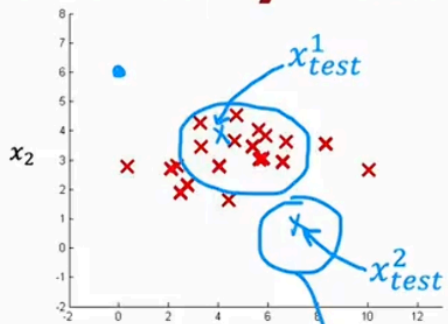
3. Given new example x , compute $p(x)$:

$$p(x) = \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2) = \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right)$$

Anomaly if $p(x) < \epsilon$

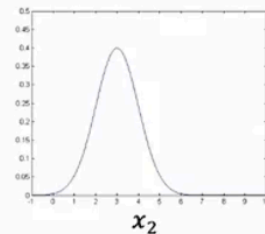


Anomaly detection example



$$\mu_1 = 5, \sigma_1 = 2$$

$$\underline{p(x_1; \mu_1, \sigma_1^2)}$$



$$\mu_2 = 3, \sigma_2 = 1$$

$$\underline{p(x_2; \mu_2, \sigma_2^2)}$$

$$\varepsilon = 0.02$$

$$p(x_{test}^{(1)}) = \underline{0.0426} \longrightarrow \text{"ok"}$$

$$p(x_{test}^{(2)}) = \underline{0.0021} \longrightarrow \text{anomaly}$$

Developing and evaluating an anomaly detection system

The importance of real-number evaluation

When developing a learning algorithm (choosing features, etc.), making decisions is much easier if we have a way of evaluating our learning algorithm.

Assume we have some labeled data, of anomalous and non-anomalous examples.

$$y = 1 \quad y = 0$$

Training set: $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ (assume normal examples/not anomalous)

$y = 0$ for all training examples

Cross validation set: $(x_{cv}^{(1)}, y_{cv}^{(1)}), \dots, (x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})})$

Test set: $(x_{test}^{(1)}, y_{test}^{(1)}), \dots, (x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$

} include a few
anomalous
examples
 $y = 1$

mostly
normal
examples
 $y = 0$

Aircraft engines monitoring example

10000 good (normal) engines

2 to 50

~~20~~ 2 flawed engines (anomalous)

$y = 1$

$y = 0$
Training set: 6000 good engines

train algorithm on training set

CV: 2000 good engines ($y = 0$)

10 anomalous ($y = 1$)

use cross validation set

tune ϵ tune x_j

Test: 2000 good engines ($y = 0$),

10 anomalous ($y = 1$)

Alternative: No test set Use if very few labeled anomalous examples

Training set: 6000 good engines 2

CV: 4000 good engines ($y = 0$), ~~20~~ 2 anomalous ($y = 1$)

tune ϵ tune x_j

Algorithm evaluation

course 2 week 3

skewed datasets

Fit model $p(x)$ on training set $x^{(1)}, x^{(2)}, \dots, x^{(m)}$

On a cross validation/test example x , predict

$$y = \begin{cases} 1 & \text{if } \boxed{p(x)} < \underline{\epsilon} \text{ (anomaly)} \\ 0 & \text{if } \underline{p(x)} \geq \epsilon \text{ (normal)} \end{cases}$$

10

2000

Possible evaluation metrics:

- True positive, false positive, false negative, true negative
- Precision/Recall
- F_1 -score

Use cross validation set to choose parameter ϵ

Anomaly detection VS. supervised learning

Anomaly detection vs. Supervised learning

Very small number of positive examples ($y = 1$). (0-20 is common).

Large number of negative ($y = 0$) examples.

$p(x)$

$y = 1$

Many different "types" of anomalies. Hard for any algorithm to learn from positive examples what the anomalies look like; future anomalies may look nothing like any of the anomalous examples we've seen so far.

Large number of positive and negative examples.

20 positive examples

Enough positive examples for algorithm to get a sense of what positive examples are like, future positive examples likely to be similar to ones in training set.

Anomaly detection vs. Supervised learning

→ Fraud detection

→ • Manufacturing - Finding new previously unseen defects in manufacturing. (e.g. aircraft engines)

- Monitoring machines in a data center

⋮

→ Email spam classification

- Manufacturing - Finding known, previously seen defects

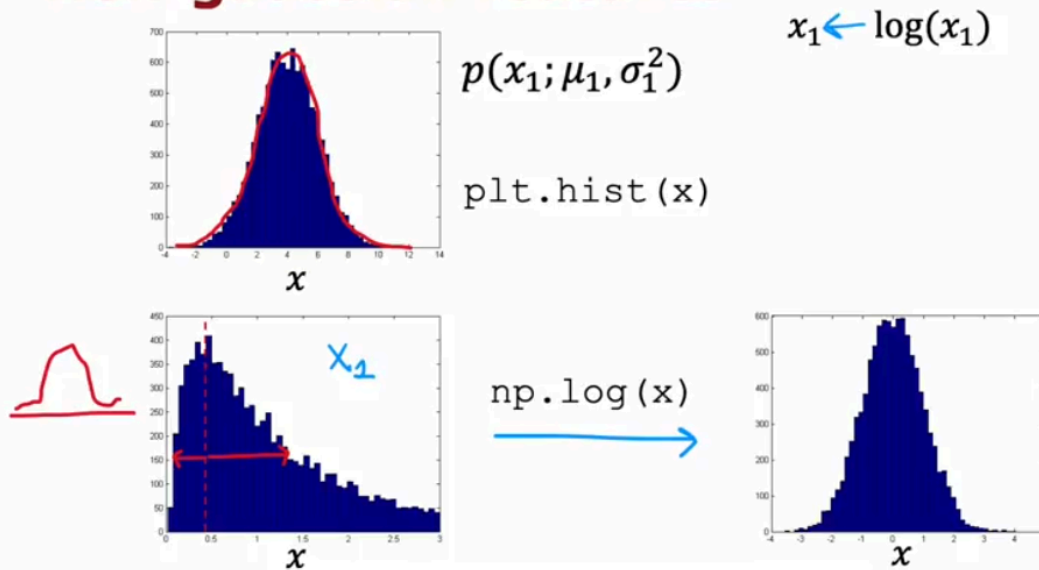
- Weather prediction (sunny/rainy/etc.)

- Diseases classification

⋮

Choosing what features to use

Non-gaussian features



Error analysis for anomaly detection

Want $p(x) \geq \epsilon$ large for normal examples x .
 $p(x) < \epsilon$ small for anomalous examples x .

Most common problem:

$p(x)$ is comparable for normal and anomalous examples.
 $(p(x) \text{ is large for both})$

